Module 2: Git Assignment - 3

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Tasks To Perform:

1. To create a Git working directory, with the following branches:
    a. Develop
    b. F1
    c. f2
2. In the master branch, commit `main.txt` file
3. Put `develop.txt` in develop branch, `f1.txt` and `f2.txt` in f1 and f2 respectively
4. Push all these branches to GitHub
5. On local delete f2 branch
6. Delete the same branch on GitHub as well

Solution steps

1. **Created the working directory**

Created directory Assignment-3

```
cd D:\intellipaat\devops\Assignments
mkdir Assignment-3
cd Assignment-3
```

```
PS D:\intellipaat\Devops\Assignments> mkdir Assignment-3


    Directory: D:\intellipaat\Devops\Assignments


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        09-11-2026     09:38                Assignment-3
```

Initialized Git inside Assignment-3 directory

git init -b master

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git init -b master
Initialized empty Git repository in D:/intellipaat/devops/Assignments/Assignment-3/.git/
```

Verified

`git status`

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git status
On branch master

No commits yet
```

Git cannot create new branches because `master` does not yet point to a commit.

2. **Created and committed main.txt in master branch**

Create main.txt in master

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> New-Item main.txt -ItemType File


    Directory: D:\intellipaat\Devops\Assignments\Assignment-3


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        09-11-2026     12:08              0 main.txt
```

Staged and committed the file main.tx

`git add main.txt`

`git commit -m "Add main.txt in master branch"`

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git add main.txt
PS D:\intellipaat\Devops\Assignments\Assignment-3> git commit -m "Add main.txt in master branch"
[master (root-commit) 4eba098] Add main.txt in master branch
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 main.txt
```

3. **Created the three branches**

`git branch Develop; git branch F1; git branch f2`

`git branch`

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git branch Develop; git branch F1; git branch f2
PS D:\intellipaat\Devops\Assignments\Assignment-3> git branch
  Develop
  F1
  f2
* master
```

4. **Added develop.txt in Develop branch, f1.txt in F1 branch and f2.txt in f2 branch**

Switched to Develop Branch

git switch Develop

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git switch Develop
Switched to branch 'Develop'
```

Created develop.txt

New-Item develop.txt -ItemType File

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> New-Item develop.txt -ItemType File


    Directory: D:\intellipaat\Devops\Assignments\Assignment-3


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        09-11-2026     20:04              0 develop.txt
```

Staged and committed the file

git add develop.txt

git commit -m "Add develop.txt in Develop branch"

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git add develop.txt
PS D:\intellipaat\Devops\Assignments\Assignment-3> git commit -m "Add develop.txt in Develop branch"
[Develop 49474da] Add develop.txt in Develop branch
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 develop.txt
```

Switched to F1 Branch

git switch F1

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git switch F1
Switched to branch 'F1'
```

Created f1.txt

New-Item f1.txt -ItemType File

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> New-Item f1.txt -ItemType File


    Directory: D:\intellipaat\Devops\Assignments\Assignment-3


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        09-11-2026     20:13              0 f1.txt
```

Staged and committed the file f1.txt in F1 branch

git add f1.txt

git commit -m "Add f1.txt in F1 branch"

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git add f1.txt
PS D:\intellipaat\Devops\Assignments\Assignment-3> git commit -m "Add f1.txt in F1 branch"
[F1 a6aca88] Add f1.txt in F1 branch
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 f1.txt
```

Switched to f2 Branch

```
git switch f2
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git switch f2
Switched to branch 'f2'
```

Created f2.txt

```
New-Item f2.txt -ItemType File
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> New-Item f2.txt -ItemType File


    Directory: D:\intellipaat\Devops\Assignments\Assignment-3


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        09-11-2026     20:22              0 f2.txt
```

Staged and committed the file

```
git add f2.txt
git commit -m "Add f2.txt in f2 branch"
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git add develop.txt
PS D:\intellipaat\Devops\Assignments\Assignment-3> git commit -m "Add develop.txt in Develop branch"
[Develop 49474da] Add develop.txt in Develop branch
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 develop.txt
```

Verification

```
git log --all --oneline --graph --decorate
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git log --all --oneline --graph --decorate
* 7e5ce56 (HEAD -> f2) Add f2.txt in F2 branch
| * a6aca88 (F1) Add f1.txt in F1 branch
|/
| * 49474da (Develop) Add develop.txt in Develop branch
|/
* 4eba098 (master) Add main.txt in master branch
```

5. **Pushed all branches to GitHub**

created a new empty GitHub repository named `Git-Assignment-3`

```
gh repo create Git-Assignment-3 --public --source=. --remote=origin
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> gh repo create Git-Assignment-3 --public --source=.
--remote=origin
✓ Created repository mg-ishaque/Git-Assignment-3 on github.com
  https://github.com/mg-ishaque/Git-Assignment-3
✓ Added remote https://github.com/mg-ishaque/Git-Assignment-3.git
```

pushed each branch

```
git push -u origin master
git push -u origin Develop
git push -u origin F1
```

git push -u origin f2

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 252 bytes | 252.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/mg-ishaque/Git-Assignment-3.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git push -u origin Develop
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 289 bytes | 289.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'Develop' on GitHub by visiting:
remote:      https://github.com/mg-ishaque/Git-Assignment-3/pull/new/Develop
remote:
To https://github.com/mg-ishaque/Git-Assignment-3.git
 * [new branch]      Develop -> Develop
branch 'Develop' set up to track 'origin/Develop'.
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git push -u origin F1
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 277 bytes | 277.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'F1' on GitHub by visiting:
remote:      https://github.com/mg-ishaque/Git-Assignment-3/pull/new/F1
remote:
To https://github.com/mg-ishaque/Git-Assignment-3.git
 * [new branch]      F1 -> F1
branch 'F1' set up to track 'origin/F1'.
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git push -u origin f2
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 12 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 277 bytes | 277.00 KiB/s, done.
Total 2 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'f2' on GitHub by visiting:
remote:      https://github.com/mg-ishaque/Git-Assignment-3/pull/new/f2
remote:
To https://github.com/mg-ishaque/Git-Assignment-3.git
 * [new branch]      f2 -> f2
branch 'f2' set up to track 'origin/f2'.
```

Verification

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git branch -r
  origin/Develop
  origin/F1
  origin/f2
  origin/master
```

6. **Deleted f2 locally**

switched away from f2

```
git switch master
```

Deleted the local branch:

```
git branch -d f2
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git branch -d f2
warning: deleting branch 'f2' that has been merged to
         'refs/remotes/origin/f2', but not yet merged to HEAD
Deleted branch f2 (was 7e5ce56).
```

Verified

```
git branch
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git branch
  Develop
  F1
* master
```

7. **Deleted f2 from GitHub**

```
git push origin --delete f2
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git push origin --delete f2
To https://github.com/mg-ishaque/Git-Assignment-3.git
 - [deleted]         f2
```

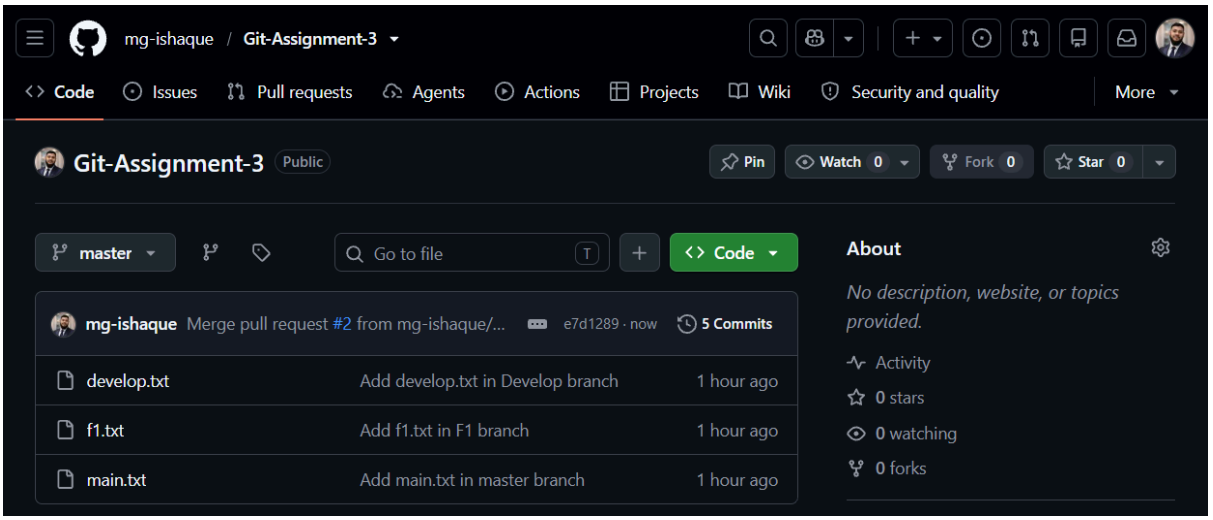
8. **Verification**

```
git fetch --prune
git branch -a
```

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git fetch --prune
PS D:\intellipaat\Devops\Assignments\Assignment-3> git branch -a
  Develop
  F1
* master
  remotes/origin/Develop
  remotes/origin/F1
  remotes/origin/HEAD -> origin/master
  remotes/origin/master
```

git log --all --oneline --graph --decorate

```
PS D:\intellipaat\Devops\Assignments\Assignment-3> git log --all --oneline --graph --decorate
* a6aca88 (origin/F1, F1) Add f1.txt in F1 branch
| * 49474da (origin/Develop, Develop) Add develop.txt in Develop branch
|/
* 4eba098 (HEAD -> master, origin/master, origin/HEAD) Add main.txt in master branch
```



List of all Commands Used

1. Directory and Git initialization

Command	Short definition
`pwd`	Display current working directory
`cd <path>`	Change directory
`mkdir Assignment-3`	Create a new directory
`git init -b master`	Initialize Git repository with `master` as the initial branch

`git status`	Show branch status and file changes
`git branch`	List local branches

2. Creating and committing files

Command	Short definition
`New-Item filename.txt -ItemType File`	Create a text file named filename.txt
`git add main.txt`	Stage the file for commit
`git add .`	Stage all changes in the current directory and below
`git commit -m "message"`	Save staged changes with a commit message
`git log --oneline`	Show condensed commit history
`git status`	Check staged, modified, and untracked files

3. Branch management

Command	Short definition

git branch branch_name	Create a branch
git switch branch_name	Switch to branch_name
git switch master	Return to master
git branch -a	List local and remote-tracking branches
git branch -r	List remote-tracking branches
git branch -d branch_name	Safely delete local f2 branch if merged
git branch -D f2	Force-delete local f2 branch

4. GitHub repository and remote management

Command	Short definition
gh repo create Git-Assignment-3 --public --source=. --remote=origin	Create a public GitHub repository and configure it as origin
git remote -v	Display configured remote repository URLs
git push -u origin branch name	Push master and configure upstream tracking

`git push origin --delete branch_name`	Delete branch_name from GitHub
`git fetch`	Download remote updates without merging
`git fetch --prune`	Fetch updates and remove stale remote-tracking references

5. Git verification and history

Command	Short definition
`git status`	Show current working tree status
`git branch`	Display local branches
`git branch -a`	Display local and remote-tracking branches
`git branch -r`	Display remote-tracking branches
`git log --oneline`	Display abbreviated commit history
`git log --all --oneline --graph --decorate`	Display a visual commit graph with all refs and branch labels